⚡ RIDEHACK'26 — IIC JUIT STARTUP CHALLENGE

SATYAMEVA AEGIS

Decentralized Trust Protocol & Academic Credential Verification Platform



Eliminating the ₹4,000 Cr academic document counterfeit black market. Transforming university credentials into tamper-proof, self-sovereign digital assets verified in under 2 seconds without accounts.

⚡ 2-Sec Instant Verify
Lifetime QR scan with institutional crest

⛓ Custom PoW Ledger
Zero gas fees + PBKDF2 dynamic salting

Gemini 2.0 Flash AI
Automated marksheet parsing & bulk CSV

🏛 JUIT / JIIT Pilot Ready
Multi-campus isolation & 100% E2E passed

TEAM: CYPHER-X (Team ID: 10) | FOUNDER: Praneeth Sai | HOST: Jaypee University of Information Technology (JUIT Waknaghat) | STAGE: Working Prototype

PROBLEM STATEMENT & MARKET PAIN

THE CRISIS IN CREDENTIAL INTEGRITY & VERIFICATION

How manual processes, paper fraud, and fragmented verification cost billions annually.

40,000+ Fake Degrees / Year

The Fake Degree Epidemic

A booming counterfeit certificate black market in India exceeds ₹4,000 Crores. Forged engineering, medical, and management degrees regularly slip undetected into the corporate workforce.

2 to 4 Weeks Turnaround

Agonizing Verification Delays

Corporate recruiters and Background Verification (BGV) agencies wait weeks for manual postal replies and phone calls to university registrars, freezing corporate onboarding pipelines.

₹500 to ₹2,000 Per Applicant

Crushing Operational Costs

Employers and background screening companies spend thousands per candidate on manual registrar checks, while universities waste massive administrative hours answering enquiries.

Zero Cryptographic Tamper Resistance

Paper & Static PDF Vulnerability

Traditional paper certificates and static PDFs offer zero cryptographic protection. Furthermore, centralized university databases remain vulnerable to rogue internal DB tampering.

NATIONAL REALITY: Over 40 Million Higher Education students in India are stuck with fragile paper certificates and zero self-sovereign digital ownership.

VALUE PROPOSITION & CORE CAPABILITIES

OUR SOLUTION: INSTANT, DECENTRALIZED TRUST

Transforming degrees into cryptographically anchored, self-sovereign digital assets.

Permanent Institutional QR

Campus-Branded Verification Anchor

High-contrast QR code embedded with the official university crest (JUIT Waknaghat / JIIT Noida) using Error Correction Level H (~30%). Lifetime-valid, permanent, and surrounded by a 360° cyber radar HUD ring.

Zero Silent Passes (Anti-Tamper)

Dynamic PBKDF2 Re-Derivation

Our verification engine doesn't just read database records; it re-derives the cryptographic hash from scratch on every request and audits the blockchain. Modifying even 1 SQL letter trips an instant TAMPERED alert.

Google Gemini 2.0 Flash AI

Intelligent Registrar Automation

Eliminates hours of manual data entry. Registrars upload certificates; Gemini AI parses name, roll number, course, and CGPA in 2 seconds. Includes atomic all-or-nothing validation for bulk CSV batches.

< 2 SECONDS

Instant Verification Speed

Reduced from 2 to 4 weeks (99.9% faster)

> 98% SAVINGS

Corporate BGV Cost Slashed

From ₹1,500 down to ₹15 per check

100% CERTAINTY

Cryptographic Tamper Detection

Zero silent passes against rogue database edits

UNDER THE HOOD: ZERO-GAS CONSENSUS & MULTI-CAMPUS SECURITY

TECHNICAL ARCHITECTURE & SECURITY ENGINE

A four-layer defense architecture guaranteeing data integrity, privacy, and speed.

Layer 1: Trust Layer

Proof-of-Work Blockchain Ledger

Custom PoW blockchain engine (SHA-256, Difficulty 4). Dual-persisted across SQLite block table and instance/blockchain.json. Zero gas fees for internal university consortium nodes.

Layer 2: Cryptographic Layer

Dynamic Salting & Zero Silent Passes

PBKDF2-HMAC-SHA256 with 100,000 iterations and per-document dynamic cryptographic salting. Full blockchain ledger traversal re-derives digests from scratch on every verification.

Layer 3: AI Document Ingestion

Google Gemini 2.0 Flash AI

Multimodal Vision AI extracts structured JSON from certificates (roll number, title, date, CGPA). Atomic all-or-nothing batch CSV validator halts invalid batches with 0 orphaned blocks.

Layer 4: Verification & UI

Permanent Crest QR & Stamped PDF

High-contrast QR with Error Correction Level H and embedded campus crest. Verification-grade Stamped PDF generator + interactive 3D WebGL ledger inspection.

PRODUCTION TECH STACK

- Backend: Python 3.14 + Flask 3.0 + SQLAlchemy + Flask-Login (Strict Role-Based Access Control)
- Intelligence & Security: Google Gemini 2.0 Flash AI · PBKDF2-HMAC-SHA256 · Pillow + QR Error Correction H
- Frontend: Responsive HTML5/CSS3 + Cyber Matrix WebGL 3D + Three.js Ledger Viewer + Desktop Native Mode
- DevOps & Scale: Docker containerized + SQLite WAL mode + Polygon L2 rollup anchoring roadmap

END-TO-END USER EXPERIENCE

LIVE PROTOTYPE: THREE ZERO-FRICTION PORTALS

Seamless user journey across University Registrar, Student, and Corporate Verifier.

🏛 PORTAL 1: REGISTRAR / ISSUER

AI-Assisted Credential Issuance

- Gemini 2.0 Flash AI auto-parses certificate PDFs into form fields in 2 seconds.
- 1-Click Blockchain Anchoring: mines block, dynamic salts, and updates ledger.
- Bulk CSV Issuance with atomic all-or-nothing pre-validation.
- Automated email notification sent to student via SMTP gateway.

🎓 PORTAL 2: STUDENT WALLET

Self-Sovereign Digital Ownership

- Permanent self-sovereign digital vault for all earned degrees and marksheets.
- 3D Interactive Card Flip: inspect underlying block index and cryptographic hash.
- Permanent Institutional QR Code embedded with official campus crest.
- 1-Click Verification-Grade Stamped PDF Certificate download.

🔍 PORTAL 3: PUBLIC VERIFIER

Instant 2-Second Verification

- Zero login barrier: employers simply scan QR or enter hash at /verify.
- Real-time cryptographic audit trail: verified against ledger in under 2 seconds.
- Anti-Tamper Engine: any altered database record trips instant RED alert.
- Corporate Batch Verify portal for background checking multiple candidates.

✅ 100% WORKING PROTOTYPE: Passed 11/11 Automated E2E Integration Suites | Dockerized | Tested at http://127.0.0.1:5000

COMMERCIAL VIABILITY & UNIT ECONOMICS

BUSINESS MODEL & MONETIZATION STRATEGY

A highly profitable, scalable B2B SaaS and pay-per-verify API revenue model.

B2B Institutional SaaS

Annual University Subscription

- Tiered annual subscription: ₹25 to ₹50 per student / year.
- University benefits: Automated Gemini AI registrar ingestion, white-labeled student wallets, official branded QR crests, and administrative analytics.
- Projected: 50 universities × 10,000 students = ₹2.0 Crores ARR.

⚡ Corporate Pay-Per-Verify API

Recruiter & BGV Screening API

- Pay-per-call API for background verification giants (HireRight, First Advantage, AuthBridge, TCS, Infosys).
- ₹15 to ₹30 per instant automated check (replacing ₹1,000+ manual checks).
- Projected: 200,000 corporate checks/year = ₹40 Lakhs high-margin revenue.

Enterprise & Consortia

State Board White-Label Deployments

- Custom white-label deployment for State Higher Education Councils and Technical University Consortia.
- Includes on-premise high-availability consortium clustering and custom ERP/LMS integrations.

UNBEATABLE UNIT ECONOMICS & PROFIT MARGINS

- Zero Gas Fees: Internal Proof-of-Work blockchain runs locally with ₹0.00 transaction fees.
- L2 Polygon Anchoring: Periodic batch Merkle-tree anchoring onto Polygon costs only ~₹0.08 per credential.
- Gross Profit Margin: >98% gross margin on corporate API verification requests.
- Customer Acquisition Strategy: Direct campus onboarding starting with Jaypee Group, expanding to Tier-1 engineering colleges.

MARKET SIZING & COMPETITIVE MATRIX

MARKET OPPORTUNITY & COMPETITIVE ADVANTAGE

Capturing India's ₹4,200 Cr credential verification market with clear technical superiority.

TAM (Total Addressable Market)

₹4,200 Crores

Indian Background Screening, Document Fraud & Academic Credentialing Market.

SAM (Serviceable Addressable Market)

₹1,100 Crores

40 Million+ students across 1,100+ Universities and 45,000+ Colleges.

SOM (Serviceable Obtainable Market)

₹45 Crores

Targeting 300+ Private & Technical Universities in Phase 1 Rollout.

Feature / Metric	Satyameva	DigiLocker	Traditional	Public Web3
Verification Speed	**< 2 Seconds**	2 - 5 Minutes	2 - 4 Weeks	15 - 45 Sec
Cost per Check	**Free / ₹15 API**	Free (Govt)	₹500 - ₹2,000	$2 - $15 Gas
Employer Friction	**Zero (No Login)**	High (OTP Auth)	High (Emails)	High (Wallet)
Internal DB Tamper Detect	**Cryptographic**	None (Trusts DB)	None	Cryptographic
AI Registrar Automation	**Gemini 2.0 Flash**	None	None (Manual)	None

EXECUTION ROADMAP & VISION

CAMPUS PILOT TRACTION & NATIONAL ROADMAP

From a Jaypee multi-campus pilot to a nationwide decentralized trust protocol.

📍 PHASE 1: CAMPUS PILOT (Q4 2026)

Jaypee Group Pilot Deployment

- Live pilot across JUIT Waknaghat, JIIT Noida, and JUET Guna.
- 15,000+ active students & 5,000+ annual graduating degrees.
- Direct integration with campus registrar & training/placement cell.
- Feedback iteration on UI and AI extraction models.

🌐 PHASE 2: STATE EXPANSION (2027)

DigiLocker & NAD Integration

- REST API integration with DigiLocker and National Academic Depository (NAD).
- Onboarding 25+ private and state technical universities.
- Corporate partnerships with top 5 Indian background verification agencies.
- Multi-tenant cloud clustering.

🚀 PHASE 3: UNIVERSAL TRUST (2028+)

National Scale & Beyond Academia

- Scaling across 40M+ Indian higher-education students under NEP 2020.
- Expanding to medical practitioner licenses, bar council credentials, and professional certifications.
- Universal decentralized digital identity infrastructure.

🎯 TRANSFORMING ACADEMIC TRUST INTO MATHEMATICAL CERTAINTY

Satyameva Aegis eliminates weeks of manual friction, protects institutional credibility, and empowers students with lifetime self-sovereign ownership. We are ready to deploy.

TEAM: CYPHER-X | FOUNDER: Praneeth Sai | RIDEHACK'26 — IIC JUIT | GitHub: github.com/praneethsai050-cmd/satyameva